Exceptional and Completed Root Kernel Calculus

A Single-Transition Problem

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Abstract

This research draft separates the unconstructed global root shriek interface from the nearby kernel descent theorem. Ordinary stalkwise inverse gluing of the distribution diagonal is impossible. The actual Verdier mate gives a dual-unit trace system with the full dualizing complex, which need not be invertible. We formulate one decisive unit–convolution–localization problem on one complete nodal support and one actual transition. The recorded local calculations and finite diagrams do not solve that problem. No global realization theorem is claimed.

1 The single-transition problem

Retain $xy = t^4$, $B = \{2\}$ and the four fixed integer-cut models of [5]. First choose the complete support $X = u^{-1}_{\{1,2\},B}(E_B)$, containing the newly introduced component and its node, and one actual neutralized root-level transition over this support. Retain finite coefficients Λ_r, independent continuous-action and distribution indices, then adic recovery and only then rationalization. No general-root statement is sought.

Problem 1.1 (A unit calculus on one complete nodal support).
Construct an exceptional/completed or mixed kernel category on this support, with actual continuous mapping spectra, an actual trace transition, a specified composition $\star$ and a unit I_X. On at least one rectangle crossing from an open branch to its node (or the reverse), construct and prove the equivalences

$$I_X \star P \simeq P, \qquad P \star I_X \simeq P,$$

including their naturality and unit identities. Retain the actual open/closed localization triangle and its connecting morphism, and prove that the specified transition and unit comparisons respect it. Neither quotienting kernel maps by their actions nor defining the category by the desired equivalences counts as a construction.

A solution must specify the finite predecessors, the variance of every transition, the order of completion, the mapping objects, and the geometric tensor/Hom and exceptional-pullback operations used to define composition. The actual bar or correspondence maps must prove both unit laws on the crossing kernel. If I_X is naturally $\Delta_*\omega_X$, its composition must be proved unital with that object. This is a different presentation until an additional comparison with ordinary shriek kernels and their unit has been established. Renaming this dual-unit does not supply such a comparison.

The ordinary-pullback obstruction is [Nearby, `cor:root-diagonal-stalk-obstruction`]: the degree-zero attaching square would require $\iota c = 1$, impossible already modulo ℓ. The actual mate is instead

$$(q \times q)_! \Delta_* \omega_Y \longrightarrow \Delta_* \omega_X.$$

The node calculation [Nearby, `lem:root-nodal-dualizing-correction`] gives $H^{-2} = \Lambda_r(1)^2$ and $H^{-1} = \Lambda_r$ on a nodal atlas. Thus an invertible scalar shift cannot identify this with the ordinary unit. These are two distinct obstacles, not missing higher homotopies.

2 Dependency boundary and retained evidence

The nearby closure, one-sided descent and Morita–Verdier realization of [5] remain independent proved manuscript statements on their specified domain. Its completed residual-gerbe unit, normalized rectangles, two coarse convolutions and carrier comparison are local inputs only. Full-fibre generators and integration remain unproved. The point-valued unit/rectangle source, ambient inertia graph objects, and actual-open geometric passage tests in the 27 September revision of [5] are three distinct domains. In particular the graph calculation is not membership in that generated source, and the two $U = \mathbf{G}_m$ tests do not supply a common strong realization. References to their diagrams below are restricted accordingly. First solve Problem 1.1; only then address full-fibre realization and strong convolution, and subsequently the four-model Verdier/refinement diamond. The diagrams recorded below are supporting material for that later stage, not a substitute for the missing kernel category or its unit laws.

References of the form [Nearby, label] identify exact statements in `verdier_kernel_skeletonization.tex`; they do not declare new assumptions or import a global root interface. The following sections preserve earlier proved local calculations and conditional diagram arguments that have been removed from the main manuscript.

3 Continuous local precursor

Use the neutralized residual groups $\Gamma_a = 0$ or $\mathbf{Z}_\ell(1)$, $G_n = \Gamma/\ell^n\Gamma$ and $\Lambda_r = \mathbf{Z}/\ell^r$.

Definition 3.1 (The continuous local kernel categories)**.** At fixed r, use the unbounded derived category of discrete continuous $\Gamma_a \times \Gamma_b$-modules over Λ_r, where Γ_a, Γ_b are either 0 or Γ. Morphisms are its full derived mapping spectra. These are actual sheaves on the relative classifying topoi; no quotient by coefficient actions is made. Put

$$A_{a,r} = C^*_{\mathrm{cts}}(\Gamma_a, \Lambda_r), \qquad I_{a,r} = C_{\mathrm{lc}}(\Gamma_a, \Lambda_r),$$

where (g, h) acts on $I_{a,r}$ by translation through $h - g$. Define the cohomological composition by

$$L \circledast_c P = R\Gamma_{\mathrm{cts}}(\Gamma_b, p_{ab}^* P \otimes^L_{\Lambda_r} p_{bc}^* L).$$

The outer group actions remain. Take the full stable idempotent closure generated by the $I_{a,r}$ and the constant rectangular kernels Λ_r on every pair, under this composition and tensor with perfect Λ_r-complexes. This definition allows the infinite-rank discrete modules $I_{a,r}$.

The notation $\circledast_c$ deliberately does not denote a root shriek operation. Continuous group cochains give its functor on actual morphisms and Fubini gives its enhanced associators.

Theorem 3.2 (Continuous diagonal and cohomological kernel comparison)**.** *The actual finite root diagonal kernels have a canonical inflation system with colimit $I_{a,r}$, which is the unit for $\circledast_c$. The functor*

$$\mathcal{R}_{c,r}(P) = R\Gamma_{\mathrm{cts}}(\Gamma_a \times \Gamma_b, P)$$

with its two cochain algebra actions is defined on full kernel mapping spectra. On the generated categories above it has canonical equivalences

$$\begin{aligned} \mathcal{R}_{c,r}(I_{a,r}) &\simeq A_{a,r}, \\ \mathcal{R}_{c,r}(\Lambda_r) &\simeq A_{a,r} \otimes_{\Lambda_r} A_{b,r}, \\ \mathcal{R}_{c,r}(P) \otimes^{L}_{A_{b,r}} \mathcal{R}_{c,r}(L) &\xrightarrow{\sim} \mathcal{R}_{c,r}(L \circledast_c P). \end{aligned}$$

The first is an equivalence of unit bimodules. *These constraints preserve units and composition on the actual mapping categories. On the integral cartesian systems generated by these kernels the comparisons pass through adic recovery and then rationalization. At the node the first comparison thereby identifies the continuous root diagonal with the regular nearby bimodule H_q.*

Proof. The representable finite diagonal of BG_n is finite etale. Its actual shriek kernel is the permutation module

$$I_{n,r} = \operatorname{Ind}_{\Delta G_n}^{G_n^2} \Lambda_r = \operatorname{Map}(G_n, \Lambda_r), \qquad \operatorname{rank}_{\Lambda_r} I_{n,r} = \ell^n.$$

For $m \geq n$ pullback of functions along $G_m \to G_n$ defines an equivariant map from the inflated $I_{n,r}$ to $I_{m,r}$. Their colimit is $I_r = C_{\mathrm{lc}}(\Gamma, \Lambda_r)$. This is the continuous coinduced module from $\Delta\Gamma$; the explicit complement to that diagonal makes coinduction exact (it is a filtered colimit of finite products). Its continuous bar Shapiro map, obtained by restriction and evaluation at the identity coset, gives

$$R\Gamma_{\mathrm{cts}}(\Gamma^2, I_r) \xrightarrow{\sim} R\Gamma_{\mathrm{cts}}(\Delta\Gamma, \Lambda_r).$$

The two projection actions of $A_{q,r}$ restrict to the two identical identity maps of $\Delta\Gamma \simeq \Gamma$. Thus this is the regular bimodule comparison, including both actions and the module map itself. It is not merely a comparison of cohomology dimensions. Equivalently compute on finite bar nerves with $I_{n,r}$ and pass to inflation: every continuous cochain with these discrete values factors through one finite level. This justifies the colimit on the actual growing kernels independently of the selected-coefficient theorem of [1].

The same Shapiro map with coefficients proves $I_b \circledast_c P = P$ and $P \circledast_c I_a = P$. Indeed the tensor with the function module is the corresponding coinduced coefficient, and evaluation at the identity coset is the unit constraint. The constant–constant composite is $A_{b,r} \otimes \Lambda_r$ on the outer group product. The complex $A_{b,r}$ is perfect over Λ_r: the two-term augmentation resolution of Λ_r over $\Lambda_r[[\Gamma_b]]$ computes it. These calculations give closure on the identity/rectangle family and then on its full thick closure, with all maps retained.

The composition comparison in the theorem is cup product after pullback to the three group factors, balanced over $A_{b,r}$, followed by continuous-cochain Fubini. For two rectangles it is the Kunneth equivalence computed by the tensor product of the finite augmentation resolutions. With an identity factor it is the specified Shapiro/unit map. Both sides are exact separately in P, L and preserve retracts, so these checks prove the comparison on their thick closures; the preceding family calculation already shows that convolution adds no other types of generators. Naturality and composition coherence are those of cup product and Shapiro on the same group bar nerves, not choices on isomorphism classes. This also proves that the map is defined on all derived kernel morphisms, without claiming full faithfulness.

The basic kernels are flat over Λ_r and their reductions are the stated basic kernels at the previous precision. Form the full finite stable/retract/convolution closure inside the category of derived cartesian systems in r. The cochain outputs of the basic kernels are the finite augmentation complexes above, which commute with derived reduction; exactness and retracts extend this to these generated systems. Thus the displayed equivalences pass

through $R\varprojlim_r$ in the derived adic category. Only then invert ℓ. The actual local star comparison of [1] identifies $C^*_{\mathrm{cts}}(\Gamma, \mathbf{Z}_\ell)[1/\ell]$ with H_q as an algebra, so the regular bimodule comparison becomes the asserted nearby unit comparison. On geometric supports this is the local node coefficient of $\Delta_* H$, not a proof of its global attachment across every model. □

Proposition 3.3 (The dual of the continuous diagonal is completed)**.** *The scalar dual of I_r is the pseudocompact module*

$$I_r^\vee = \varprojlim_n \Lambda_r[G_n],$$

with transition maps summing coefficients over quotient fibres. Continuous scalar biduality recovers I_r; ordinary discrete coefficient duality does not keep it in the same finite-coefficient class used in [1].

Proof. The dual of the pullback map of finite function modules is the quotient-fibre sum on distributions. Derived Hom takes their filtered colimit to the inverse limit of their duals. These inverse maps are surjective, so there is no higher inverse-limit term. A continuous functional on the resulting profinite module with values in the finite discrete ring factors through a finite quotient. Its dual is therefore the original colimit of functions. The pairing is evaluation, equivariant for the two contragredient actions. Recovering r gives the completed integral distribution module; rationalization must still follow this recovery. This constructs the required local dual pair, but does not supply a global six-operation theory mixing all discrete and completed kernels. □

4 Coefficient branch and later coherence material

4.1 An actual branch calculation and its dual boundary

The following calculation supplies a coefficient branch face of the missing four-model gluing diagram. It deliberately uses the normal-pair Kummer map, not central-gerbe purity.

Lemma 4.1 (Continuous Kummer link and its trace-dual boundary)**.** *Fix one branch coordinate x in one of the four models. On its strictly local smooth branch let j_n be the punctured inclusion and i_n the closed inclusion in the Kummer chart $x = z_n^{\ell^n}$. Set $L_{n,r} = i_n^* R j_{n,*} \Lambda_r$, with its actual deck action. The pullback comparisons for $z_n = z_m^{\ell^{m-n}}$ give*

$$\operatorname*{hocolim}_n L_{n,r} \simeq E_r$$

as a discrete continuous inertia object, through the actual unit. For a selected unipotent branch coefficient V, the corresponding continuous link is its continuous inertia module, with the sign of the specified branch coordinate. These are natural comparisons on the selected mapping category. Dualizing the finite localization diagrams gives their actual trace-dual inverse diagrams and connecting arrows.

Proof. The punctured strict branch has cohomology Λ_r in degree zero and $\Lambda_r(-1)$ in degree one, and no higher cohomology. Kummer functoriality sends the degree-one class of z_n to ℓ^{m-n} times that of z_m; it acts identically on degree zero. These statements identify the maps of the actual link complexes, not an arbitrarily selected splitting. The maps are induced by the same Kummer morphisms as the deck actions. At fixed r the positive-degree system is zero after sufficiently many transitions. Filtered colimits in discrete continuous modules are exact, so the actual unit $E_r \to \operatorname{hocolim}_n L_{n,r}$ is an equivalence; its degree-zero action is the trivial action. For a selected coefficient use its finite-action presentation and pull back to a Kummer level trivializing it. The same calculation with that coefficient, together with

naturality of the unit, gives the stated inertia module. Alternatively exactness extends this natural comparison from the unit to its thick closure. The two branch coordinates may be inverse; their induced inertia map, rather than a chosen identification of cohomology lines, records that sign.

At each level retain the entire localization triangle and its connecting arrow:

$$j_{n,!}V_n \longrightarrow Rj_{n,*}V_n \longrightarrow i_{n,*}L_{n,r}(V) \xrightarrow{\delta_n} j_{n,!}V_n[1].$$

Open pullback and the adjunction comparison for Rj_* give the maps of these actual triangles; the connecting squares are obtained from their cofiber diagrams. Apply geometric Verdier duality to those finite diagrams, retaining the smooth-branch dualizing factor $\Lambda_r(1)[2]$ on the open side and point duality on the closed side. This gives the inverse diagrams of the actual adjunction mates and their trace boundaries. In particular on the closed link they satisfy

$$R\varprojlim_n D_{\Lambda_r} L_{n,r}(V) \simeq D_{\Lambda_r} \operatorname*{hocolim}_n L_{n,r}(V),$$

with the evaluation pairing. For a constant coefficient the dual of the degree-one transition is again multiplication by ℓ^{m-n} and is pro-zero at fixed r; the surviving degree-zero pairing is ordinary evaluation. Finite source Verdier factors are retained in the boundary diagram; this scalar link identity does not remove them. No equality $q^! = q^*$ for arbitrary coefficients on a ramified normal pair is used. □

The lemma compares coefficient links and their dual boundaries for the fixed branch charts. It does not compute the product links of the completed diagonal, normalized rectangles, or contraction graph kernels. Those product-link maps, involving distribution limits as well as the continuous-action index, are required before a global kernel functor can be glued. In particular a coefficient branch comparison cannot be used as a substitute for those kernel maps.

4.2 One actual diagram for the four compatibility checks

We organize the remaining root comparisons by one diagram, with a precise distinction between identities of assembled diagrams and invertibility of their interpretation as global kernel operations. This avoids assuming the desired coherence in a definition of the root source.

At precision r, retain the actual finite neutralized root charts of the four fixed models, their support products, their diagonal and graph correspondences, and their finite Kummer branch charts. For an expression in already specified finite coefficients, form its relative bar and Kummer Cech diagrams on these actual products. Include the open/closed localization arrows, the chosen carrier projection, the two refinement paths, and the evaluation arrows. Call this diagram $\mathcal{B}_r$. Its maps are the actual pullback, shriek direct image, units, counits and their Beck–Chevalley mates in the fixed enhancement. In particular the nonrepresentable ! leg is not replaced by its $*$ leg. No unspecified global normalized coefficient is inserted into $\mathcal{B}_r$.

There is a distribution-presentation index $\mathbf{n}$ and a continuous-action index $\mathbf{m}$ dominating it, one entry for each inertia factor. On discrete continuous mapping/cochain faces the order is $\operatorname{hocolim}_{\mathbf{m}}$ followed by $R\varprojlim_{\mathbf{n}}$. On their trace-dual faces use the dual order prescribed by evaluation. Derived precision recovery is last. Equal-variance indices may be synchronized by cofinality; a direct and an inverse index may not be synchronized this way.

Proposition 4.2 (Four faces of the same assembled diagram)**.** *On the local completed kernel source and the coefficient branch faces constructed above, the following comparisons come from $\mathcal{B}_r$, and are compatible wherever their operations have already been identified:*

1. *units and convolution substitution;*

2. *localization/cofiber connecting arrows;*

3. *evaluation, Verdier mates and carrier traces;*

4. *the actual refinement/base-change comparisons and their composition simplices.*

Their common higher compatibility is inherited from the same bar, adjunction and trace diagram. For a global four-model root kernel realization, this assertion becomes applicable only after the product-link and completion comparison maps have been constructed and proved invertible; it does not assert those missing maps.

Proof. For a string of actual correspondences, evaluate its coefficient on the full product and use the multiple relative bar. A substitution deletes an intermediate bar by its augmentation; inserting the diagonal gives its degeneracy and regular unit. Whiskering two deletions in either order is the same two-dimensional bar face. This supplies the first comparison and its associativity cells. In the local completed module model these are exactly the evaluations used in Theorem [Nearby, `thm:completed-root-shriek`].

Retain an open/closed arrow in that same diagram instead of collapsing it to its object values. Its cofiber vertex and the connecting arrow are then functorially determined. Applying a natural comparison $\tau : F \to G$ gives $G(\delta)\tau_{Q_2} = \tau_{Q_0}[1]F(\delta)$, with its specified homotopy, for the boundary $\delta : Q_2 \to Q_0[1]$. Finite cones and retracts, and every limit or colimit functor used in the assembly, are exact. Thus this identity survives assembly on the specified diagrams. Lemma 4.1 identifies the actual single-support branch boundary face; no identification of a product-kernel boundary follows just from it.

Taking a mate means applying the given adjunctions to this same face. Its evaluation satisfies, for an actual finite refinement,

$$e_n \circ (t_{m,n} \otimes 1) = e_m \circ (1 \otimes \inf_{n,m}).$$

For composable structural maps the final pairing uses $\mathrm{Tr}_{gf} = \mathrm{Tr}_g \circ Rg_! \,\mathrm{Tr}_f$. These are faces of the finite six-operation diagram, including ! on nonproper projections. The carrier comparison is precisely this pairing followed by the source-augmentation evaluation of Theorem [Nearby, `thm:selected-carrier-interface`]. Consequently taking mates, inserting a unit and contracting another kernel give the same evaluation diagram. Biduality is used on selected perfect outputs or on the specified discrete/pseudocompact dual pair, not on an unrestricted completed sheaf class.

Finally the refinement simplex is the composition simplex of the actual chart maps and their adjunctions. In local graph coordinates it is $R_a \widehat{\otimes}_{R_b} R_b \to R_a$ and its iterated substitution, as in Proposition [Nearby, `prop:completed-root-graphs`]. Its base-change faces are the prescribed Beck–Chevalley maps, not arbitrary identifications chosen after comparing stalks. Taking a mate of this simplex or inserting the localization arrow therefore gives the same simplex in the larger diagram.

Apply the assembly functors to $\mathcal{B}_r$ *as a diagram.* A functor carries simplices and their substitutions to simplices; thus all four identities and their mixed cells remain identities. This argument does not say that assembly commutes with an arbitrary relative tensor or geometric direct image. Those are additional comparison maps. They are proved for the local unit/rectangle words and coefficient faces already used here. For the still missing product links they have not been proved. Once an actual comparison of diagrams has invertible components there as well, restriction to the full generated source will retain the common cells without four separate choices of coherence data. □

Lemma 4.3 (Independent rectangular product-link faces)**.** *For two fixed branch charts and their punctured inclusions j_x, j_y, use the product-open link*

$$(i_x \times i_y)^* R(j_x \times j_y)_*(V \boxtimes W).$$

This is the link to the product of the two punctured branches, not the link to the full punctured neighbourhood of the corner. Exterior products of actual finite, underlying-perfect selected coefficients have this product-link comparison given by the tensor product of the two branch bar diagrams. Their two restriction orders agree, including the localization-arrow faces. Continuous-action colimits and the dual trace diagrams retain this comparison. With distribution coefficients the same statement holds with independent presentation indices and the completed exterior product, defined by their inverse system; it does not exchange a distribution inverse limit with a continuous-action colimit.

Proof. At fixed coefficient presentations the product Kummer torsor is the product of the two actual torsors. Its two Cech directions form a bisimplicial diagram. The scalar product-link map is the external-product/Kunneth comparison on those nerves. The branch cohomology has finite amplitude and the coefficients here are underlying-perfect, so totalization of this finite-dimensional branch calculation gives the comparison with the product link. The two orders are identified by the same bisimplicial Fubini map, with the Koszul signs; this proves equality of the specified maps. Tensor with, or restrict, the actual open/closed arrow before this calculation to obtain the localization faces of the same diagram.

Take the continuous-action colimits at fixed presentations first. Their tensor comparison is valid since derived tensor preserves these colimits separately. On the dual trace diagrams use derived Hom of this equality, with their finite evaluation pairings. For distribution coefficients retain the two presentation indices and then take their inverse limit; this is the completed exterior product, not an uncompleted tensor product of the two limits. This proves precisely the claimed comparison. It applies to independent exterior factors and this product-open face. Other corner faces still have to be retained in a full gluing diagram. A diagonal or graph imposes a relation between the two inertia directions and is not such a factorization. Nor does this lemma construct the missing global normalized source factor on a whole refined fibre. □

5 Finite unit and dual-unit diagrams

Lemma 5.1 (Unit and dual-unit diagrams on the complete supports)**.** *At finite root precision and level, the preceding adjunction construction applies to every actual transition $q : Y \to X$ between the fixed support charts, including charts meeting more than one branch. With $F_X = \Delta_{X,!}\Lambda_r$ it gives*

$$(q \times q)^* F_X \longrightarrow F_Y, \qquad (q \times q)_! \Delta_{Y,*}\omega_Y \longrightarrow \Delta_{X,*}\omega_X, \quad \omega_X = D_X\Lambda_r.$$

Composition, restriction of the localization arrow, and the Verdier-mate identities belong to the same actual adjunction diagram. These maps are not asserted invertible. In particular, a refinement does not identify its fibre relation with its diagonal. The second family is a family of dual-unit *trace maps; on a singular support it is not a family of shriek-unit maps obtained by cancelling one constant twist and shift.*

Proof. The construction of β in Proposition [Nearby, `prop:root-diagonal-branch-transition`] only uses the actual diagonal counit and composition of pullbacks. It does not use smoothness. Diagonals of the finite separated root charts here are proper and representable. Proper duality therefore identifies $DF_X = \Delta_{X,*}\omega_X$, giving the second family

by taking the mate of the first. The unit/counit identities for three composable maps identify the two resulting β maps. Their dual identity is trace transitivity. Retaining the open/closed arrow and its cofiber in this natural transformation gives its localization and boundary faces. This is a particular actual instance of the diagram in Proposition 4.2; it requires no choice of a new comparison on each branch. For a refinement, the first source still has support on the full inverse image of the diagonal. Nothing in the counit construction asserts that this support equals the fine diagonal or that the counit is an equivalence. Finally proper duality gives ω_X itself, not an invertible replacement for it. □

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